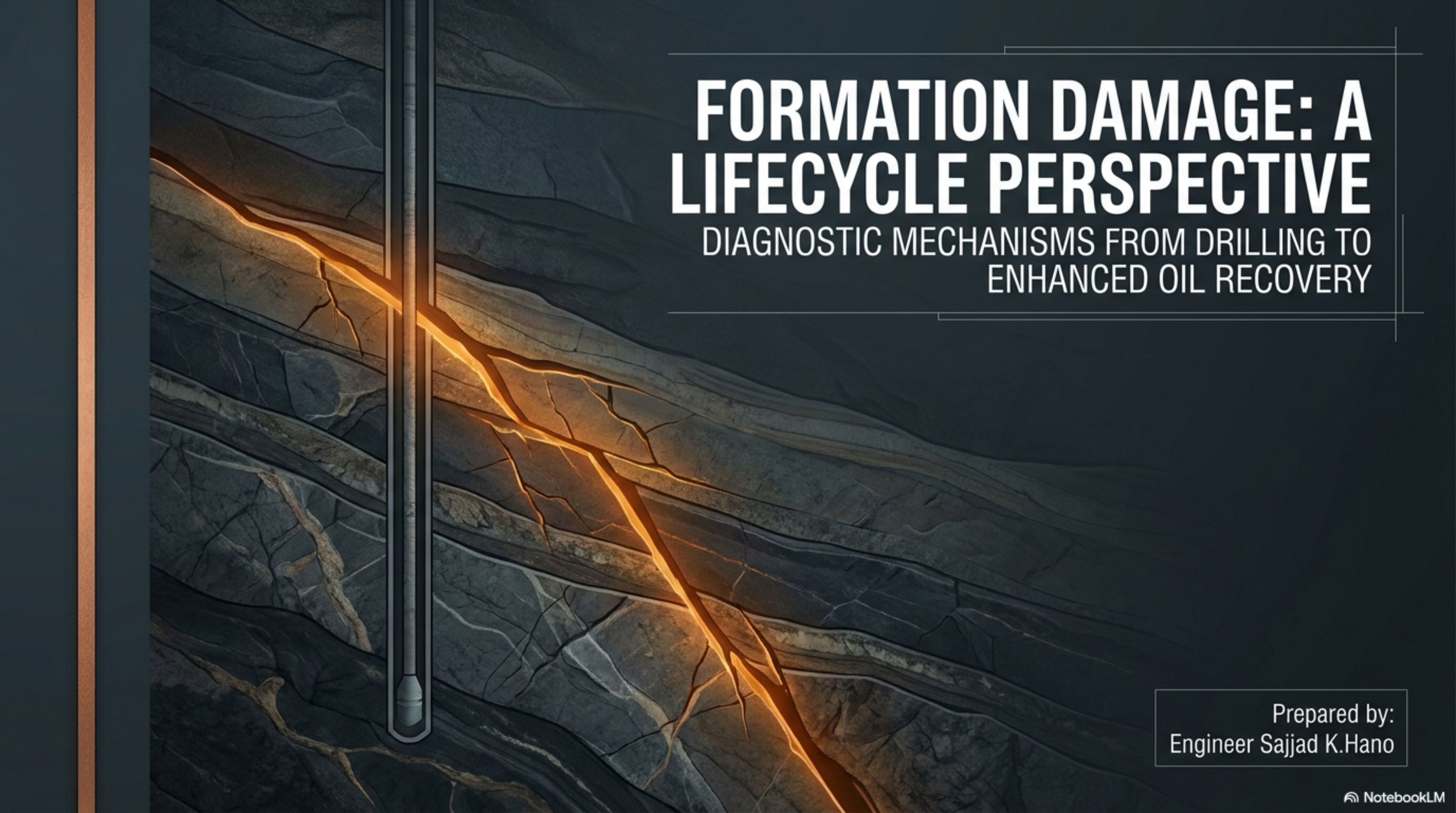




FORMATION DAMAGE: A LIFECYCLE PERSPECTIVE

DIAGNOSTIC MECHANISMS FROM DRILLING TO
ENHANCED OIL RECOVERY

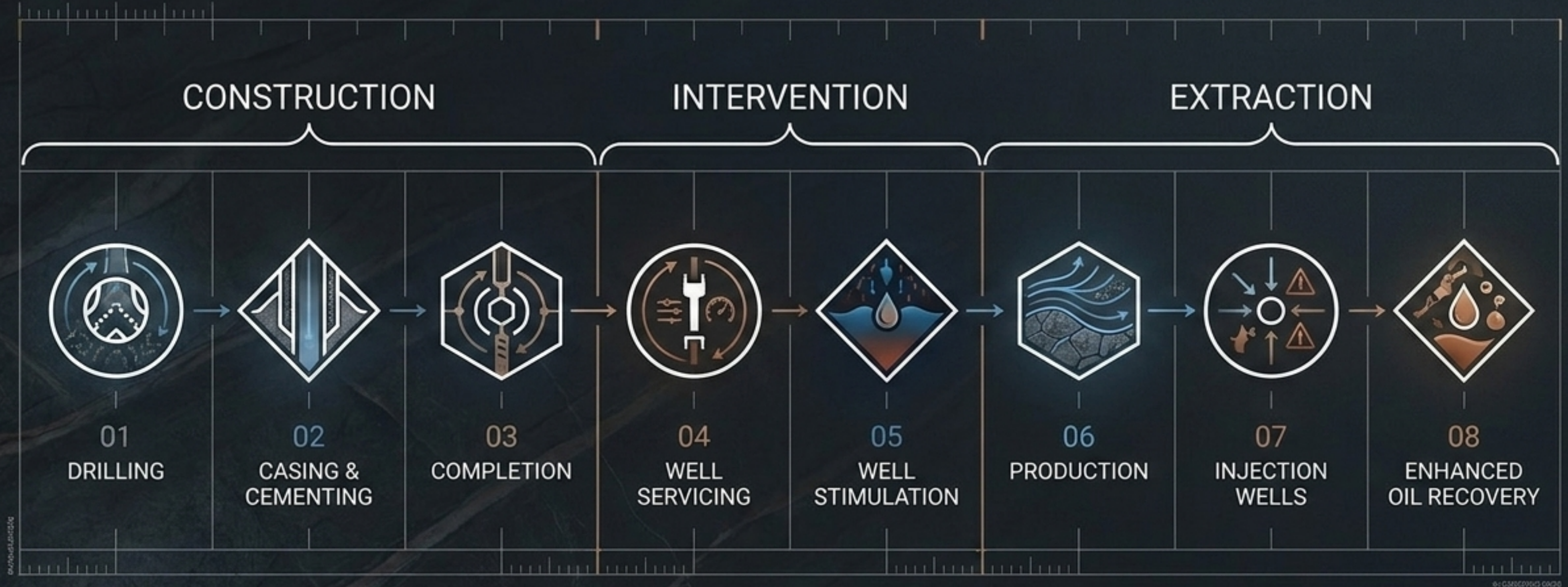
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THE FUNDAMENTAL THREAT TO PERMEABILITY

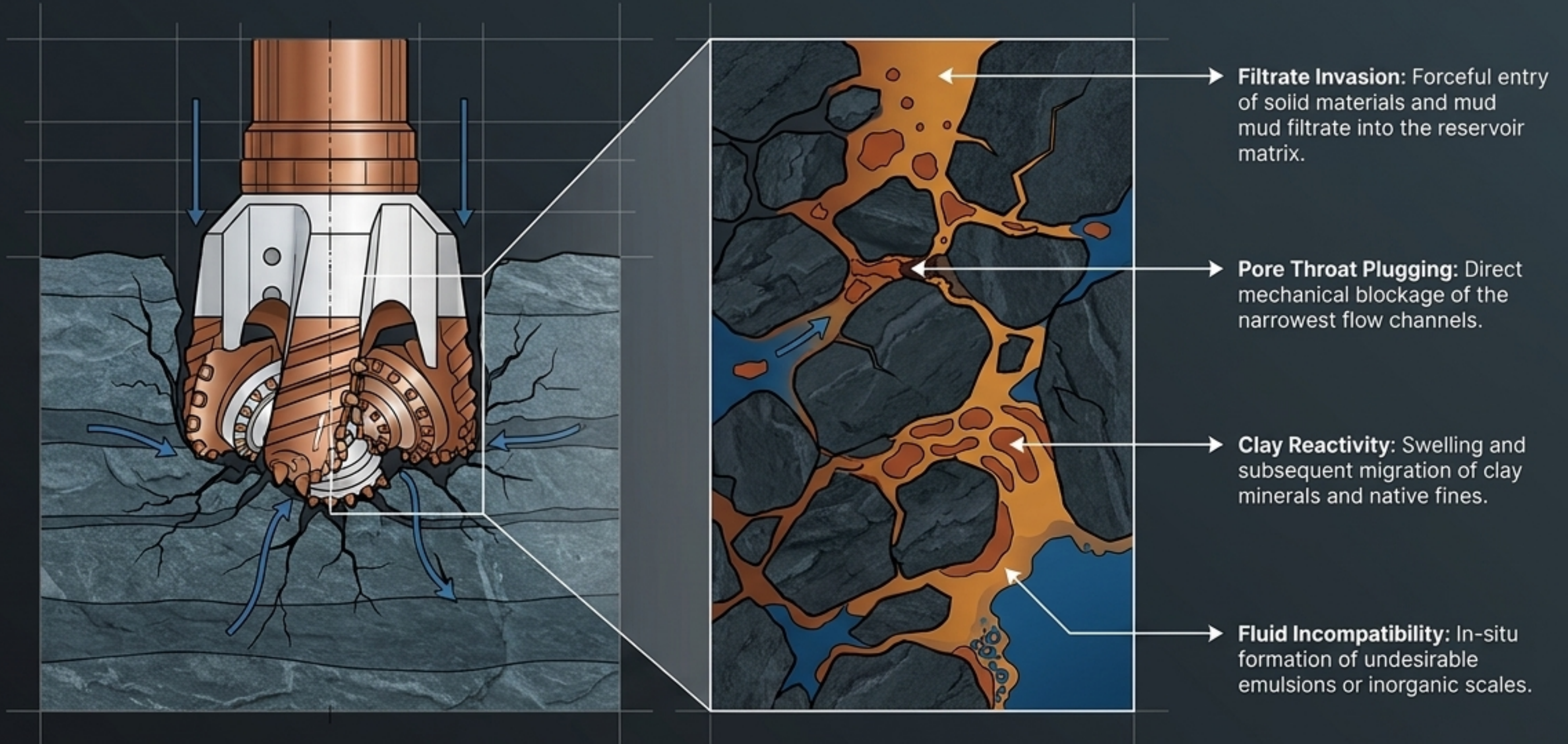
Formation damage is an operational inevitability. Throughout a well's lifecycle, every intervention—from the drill bit to EOR fluids—threatens the natural flow capacity of the reservoir. Understanding these mechanisms is the first step to mitigating them.



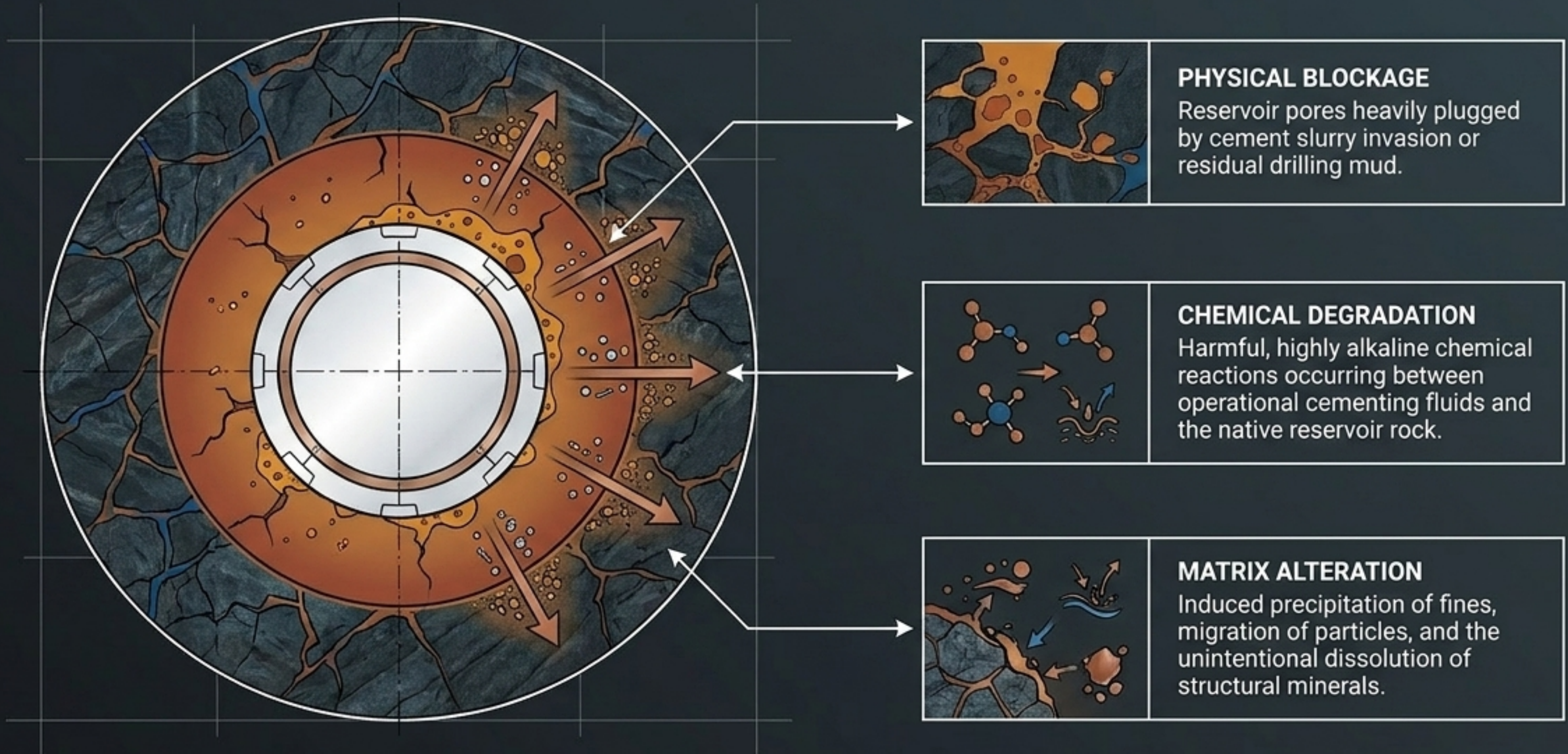
THE 8-STAGE VULNERABILITY TIMELINE



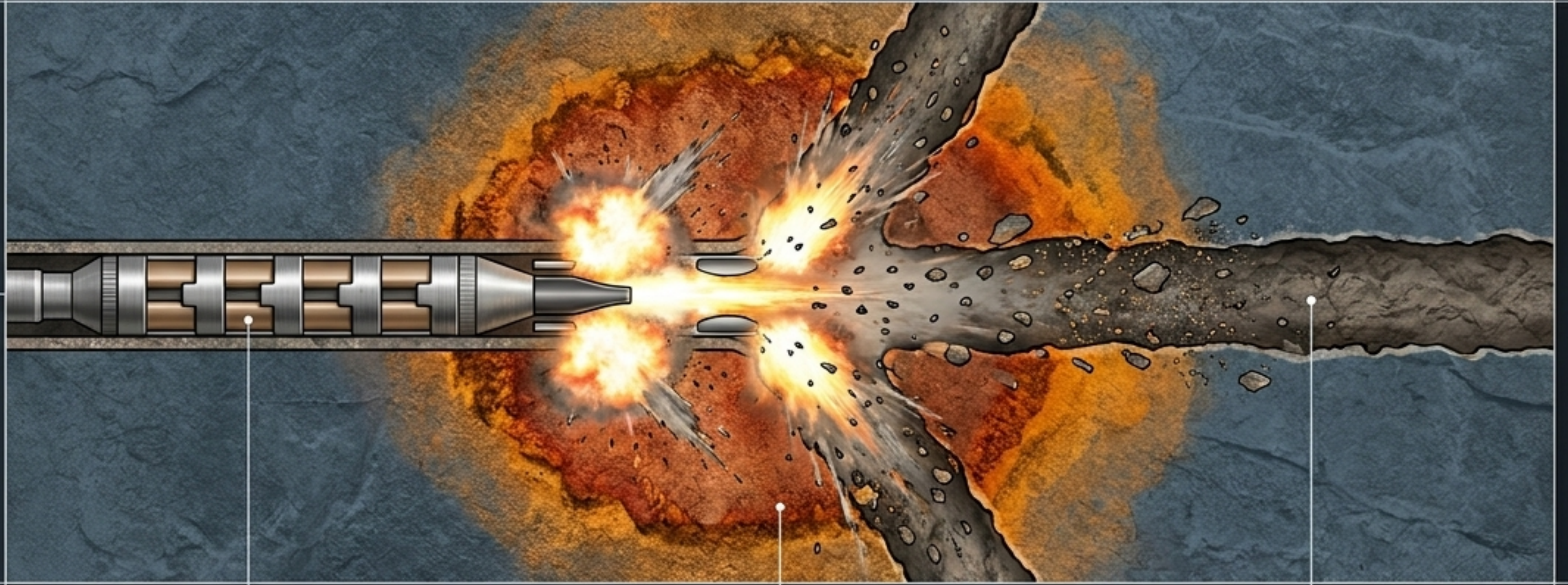
01 / DRILLING OPERATIONS



02 / CASING & CEMENTING



03 / COMPLETION MECHANICS



OVERBALANCE INVASION:

Unintended entry of completion fluids and solids into the formation driven by high operational pressures.

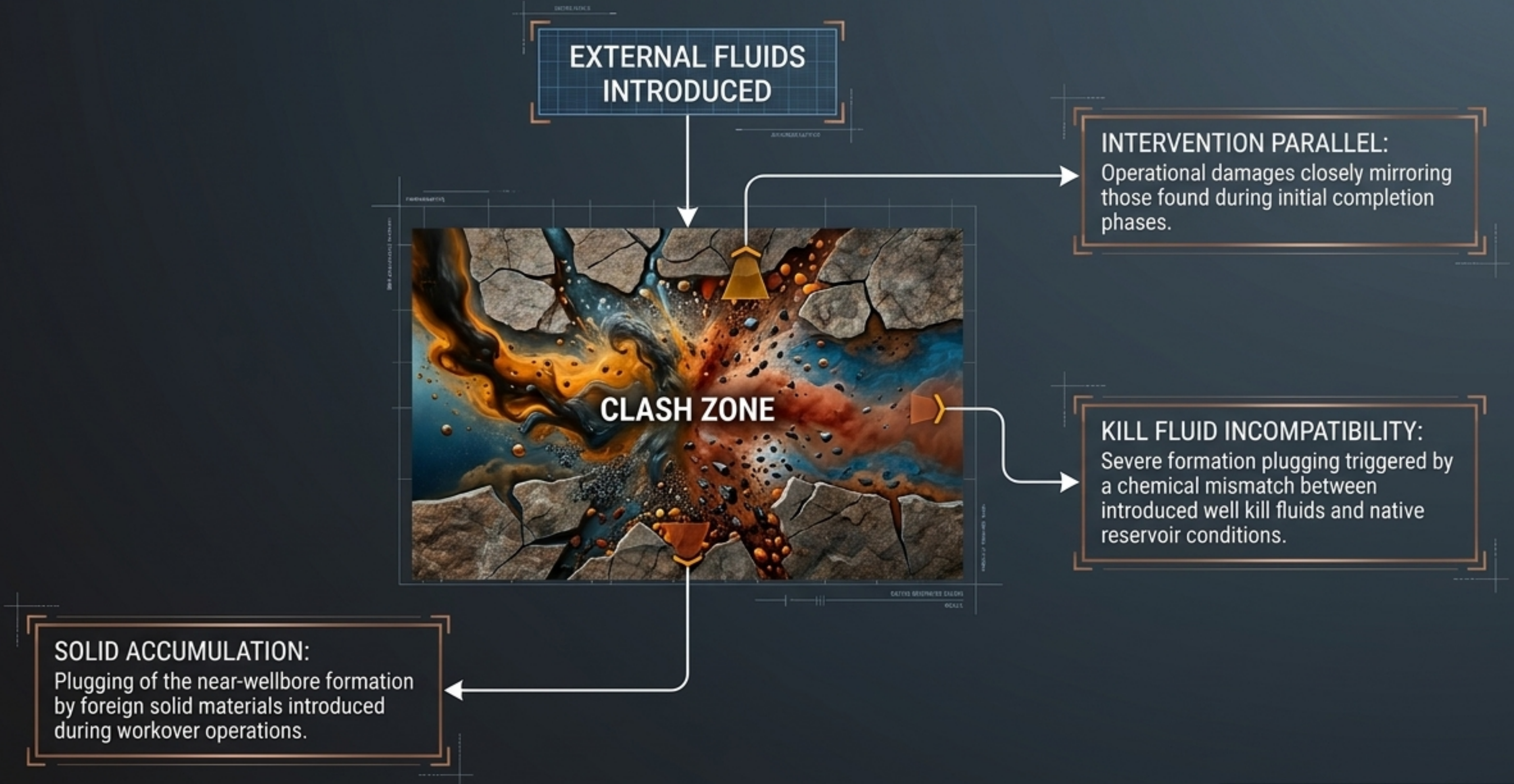
EXPLOSIVE DAMAGE:

Severe rock compaction and debris accumulation resulting from perforation explosives.

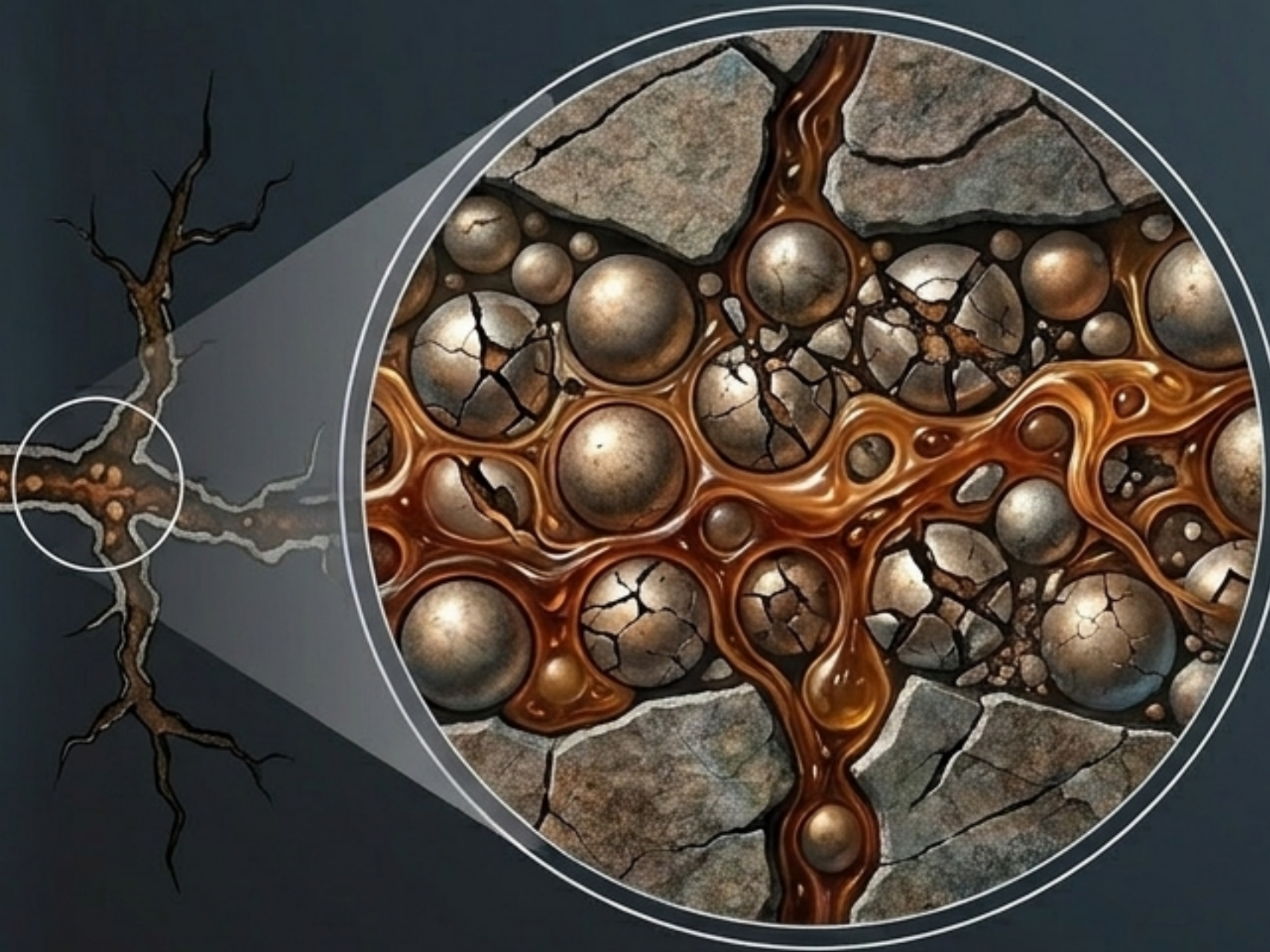
TUNNEL PLUGGING & WETTABILITY:

Physical blockage of the newly created perforation tunnels alongside detrimental alterations to localized rock wettability.

04 / WELL SERVICING INTERVENTIONS



05 / WELL STIMULATION SIDE-EFFECTS



RESIDUE PLUGGING:

Critical pores and induced fractures blocked by fracturing fluid residues or external solid additives.

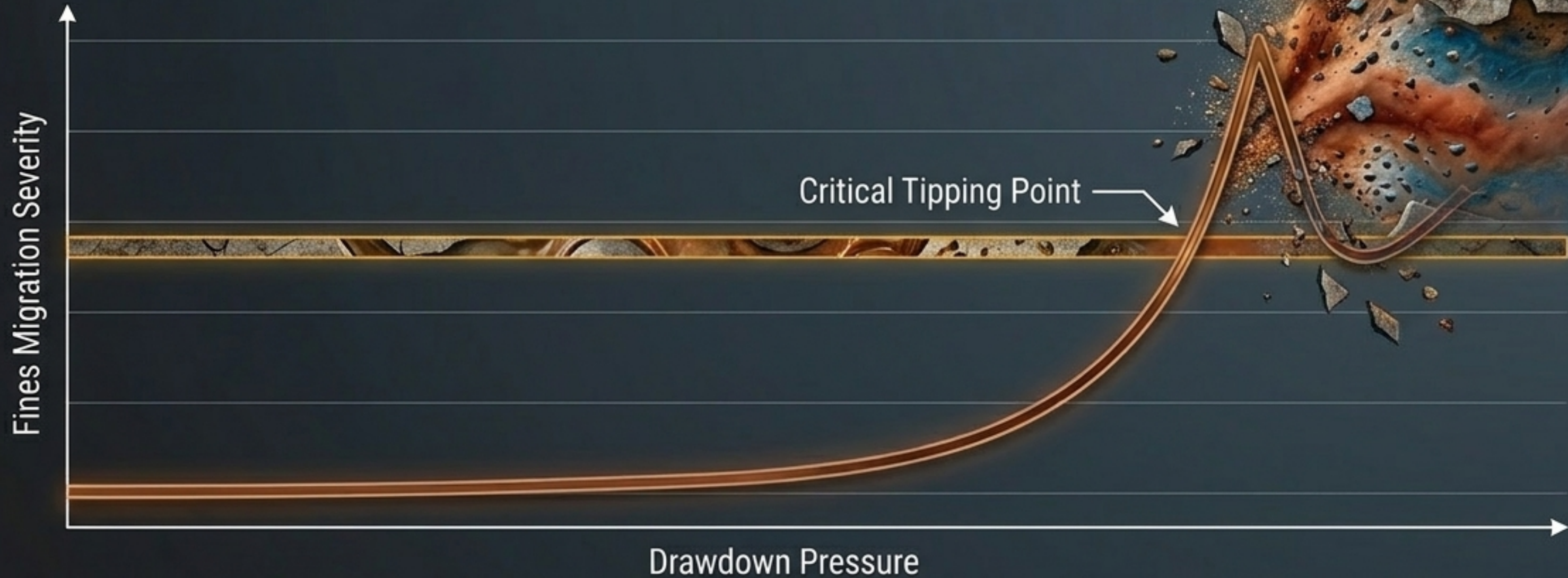
REACTION BYPRODUCTS:

Fines migration and secondary precipitation resulting directly from acid or chemical reactions.

CONDUCTIVITY LOSS:

Deterioration of proppant efficiency (crushing/embedment) leading to a direct decrease in overall fracture conductivity.

06 / PRODUCTION DYNAMICS



Excessive Drawdown:

High-velocity fluid movement resulting from excessive pull, triggering aggressive movement and bridging of fine materials.

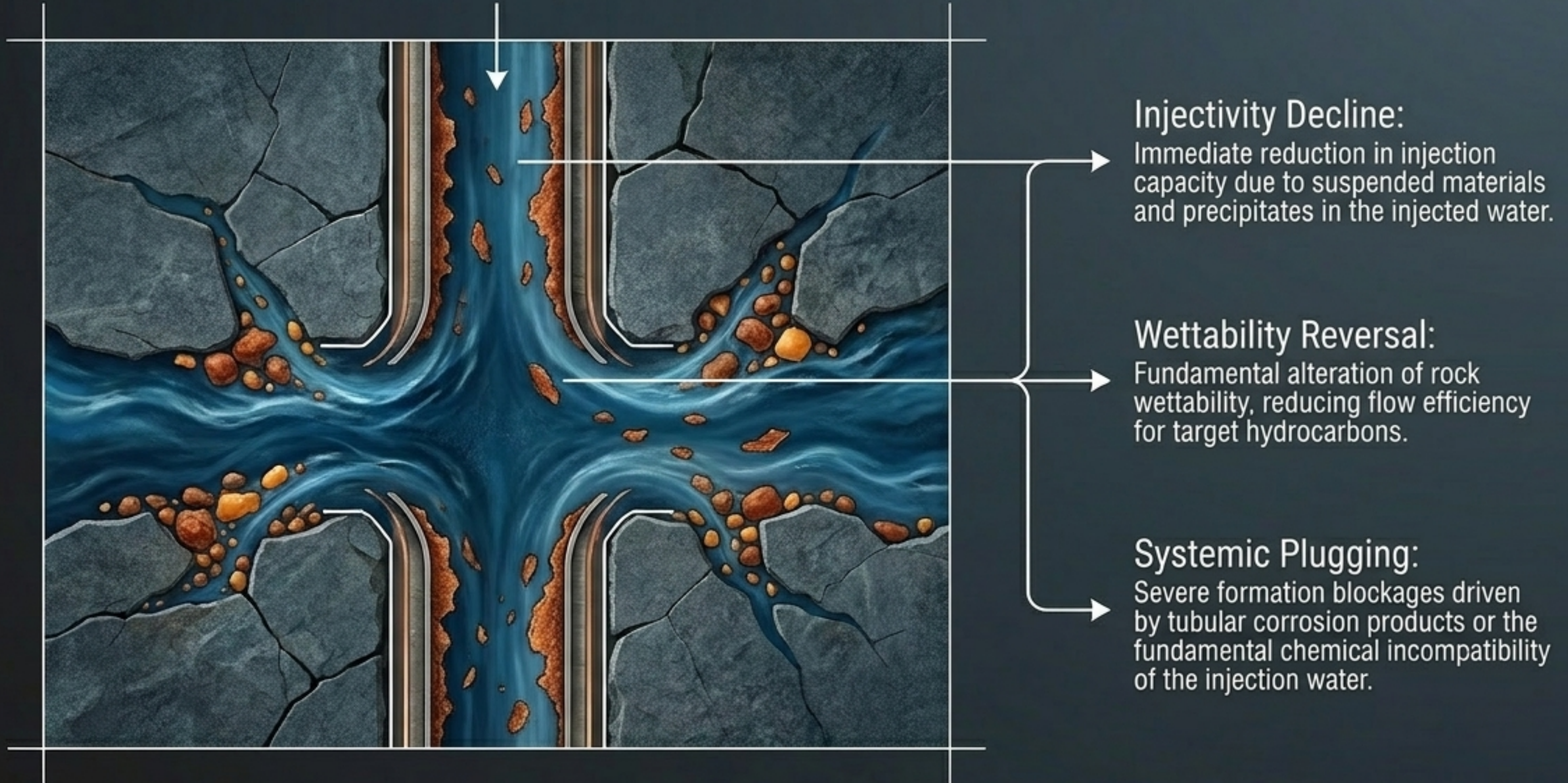
Scale Deposition:

The continuous thermodynamic formation of both organic (asphaltenes/waxes) and inorganic (carbonates/sulfates) precipitates.

Sand Failure:

Uncontrolled sand production leading to the catastrophic plugging and failure of downhole sand control systems.

07 / INJECTION WELLS & SECONDARY RECOVERY



08 / ENHANCED OIL RECOVERY (EOR)



Chemical Sensitivity:

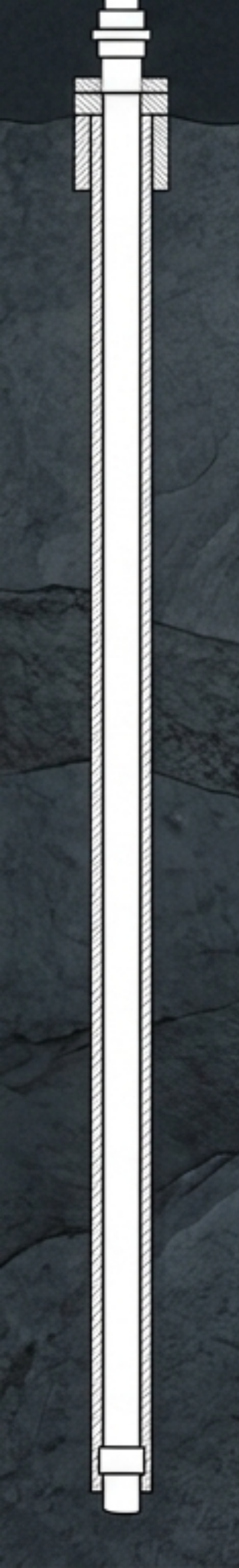
Severe clay swelling and sustained migration of fine materials triggered by chemical flood agents.

Heavy Deposition:

Massive precipitation of complex carbonates and heavy asphaltenes induced by changing reservoir thermodynamics.

Emulsion Blocking:

The formation of highly viscous, trapped emulsions causing drastic and permanent reductions in formation permeability.



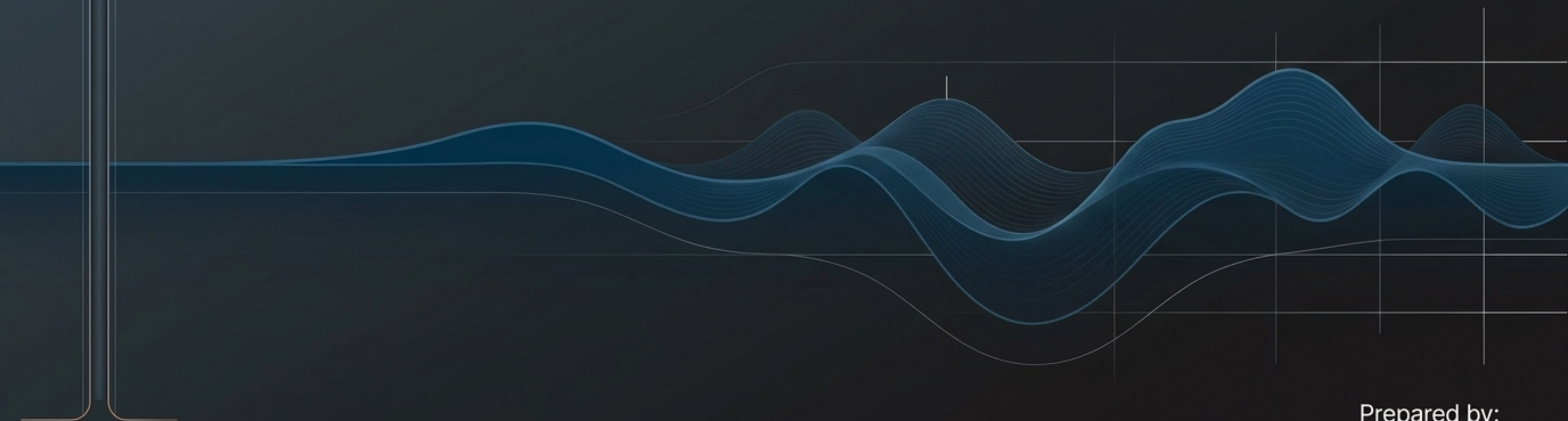
The Architecture of Damage: Categorical Synthesis

Tracing the root causes of permeability loss across mechanical, chemical, and fluid-dynamic domains allows for preemptive operational design rather than reactive treatment.

	Mechanical (Plugging/Solids)	Chemical (Incompatibilities/Scales)	Fluid-Rock (Wettability/Emulsions)
01 Drilling	●		●
02 Casing & Cementing	●	●	
03 Completion	●	●	●
04 Well Servicing	●		
05 Well Stimulation		●	●
06 Production		●	●
07 Injection Wells	●	●	●
08 EOR		●	●

Preserving Reservoir Potential

Formation damage is a continuous, evolving threat from the first day of drilling to the final days of EOR. By anticipating chemical, mechanical, and fluid incompatibilities at every operational stage, we can optimize well productivity, protect permeability, and significantly extend asset lifespan.

A decorative graphic on the left side of the slide. It features a vertical line representing a wellbore, with a horizontal line extending from it. Below this, there are several wavy, horizontal lines in shades of blue and teal, creating a sense of movement or fluid flow. The background is a dark, textured blue.

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